FACTORS AFFECT FOR THE HOUSING PRICES IN AMES,LOWA,USA

by
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DECLARATION

I, D.L.C.K.Dasanayaka , declare that the presented project report titled, "FACTORS AFFECT FOR THE HOUSING PRICES IN AMES,LOWA,USA" is uniquely prepared by me based on the group project carried out under the supervision of Dr. K.D.Prasangika Department of Mathematics, Faculty of Science, University of Ruhuna, as a partial fulfillment of the requirements of the level II Case Study-I course unit (MIS2231) of the BSc(Honors)Financial Mathematics and Industrial Statistic in Department of Mathematics, Faculty of Science, University of Ruhuna, Sri Lanka. It has not been submitted to any other institution or study program by me for any other purpose.

Signature:.....

Date:.....

SUPERVISOR'S RECOMMENDATION

I/We certify that this study was carried out by D.L.C.K.Dasanayaka under my/our supervision.

Signature:.....

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ABSTRACT

This study investigated the key determinants of residential housing prices using a multiple linear regression model on a dataset of 1335 housing transactions.It mainly focus on how log lot area,Overall quality,Year built,total basement surface area,Ground living area,Total rooms above ground and Garage area affect on sales prices of houses. The primary objective is to develop a highly predictive model by identifying a significant subset of variables influencing housing sales prices.To achieve this, the research utilized the backward elimination method using R software.

Keywords: Multiple linear regression,Backward elimination method,sales price

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CHAPTER 1
Introduction

Overview

This chapter is organized as follows. First focuses on giving an understanding of the housing prices and key steps used in the process of developing the model. In later research problem, research questions and objectives of this research will be defined.

1.1 Background of the study

The real estate market is a massive and dynamic market that is affected by various factors, and accurate valuation of property is a challenging but rewarding exercise for many stakeholders such as buyers, sellers, investors, and property professionals. Familiarity with the determinants of house prices is important to enable informed decisions, analyze trends in the market, and make economic forecasts. The worth of a property is not established by where it is but by a combination of structural features, quality levels, size, age, and other common characteristics of a building.This study will investigate the impact of various house characteristics such as lot area, style of house, overall quality, year of construction and refurbishment, dimensions of basement and living area, quality of heating, number of bedrooms and bathrooms, number of rooms in total, number of fireplaces, and garage space on the sale price of houses. The study will be conducted based on the Ames Housing data set collected by Kaggle website.

1.2 Problem Statement

Accurate house sale price forecasting is a constant challenge in the property market due to the interplay among numerous factors that decide the value of a property. While intonation suggests common sense dictates that certain aspects are more powerful, trying to gauge their definitive influence and choose which ones are most important is a tricky task. In this research, we intend to address the problem of how to comprehend and quantify the relationship between various structural and quality attributes of a house and its ultimate selling price.At the end of study, we hope to better understand factors that influence house sale prices, which can lead to better market insights.

1.3 Significance of the study

This study is of significant importance for a variety of reasons. It aims to explore the relationships between various features of housing and Sale Price. The findings will help establish those parameters that have the greatest impact on house pricing, and these attributes can be highly beneficial to prospective homeowners in making effective buying decisions, sellers in setting optimal prices, and real estate agents in making realistic market estimates. Moreover, such knowledge will enable property developers to invest more in those attributes that contribute most to value, and urban planners to predict the economic impacts of different housing qualities. Overall, this research will assist in building a smarter and more efficient real estate market by exposing effective approaches to comprehend and influence property values.

1.4 Research Questions

- What is the relationship between log lot area(Logarithm of lot area) and Sale-Price?
- What is the relationship between Overall Qual (Overall quality rating) and Sale-Price?
- What is the relationship between Year Built and Sale-Price?
- What is the relationship between Total Bsmt SF (Total basement square footage) and Sale-Price?
- What is the relationship between Gr Liv Area (Above grade living area square footage) and Sale-Price?
- What is the relationship between TotRms AbvGrd (Total rooms above grade) and Sale-Price?
- What is the relationship between Garage Area (Size of garage in square feet) and Sale-Price?
- Which independent variables are the most significant predictors of Sale-Price in the dataset?

1.5 Hypothesis

H_0: Null Hypothesis

There is no relationship between dependent variable and independent variables.

H_1: Alternative Hypothesis

There is a linear relationship between dependent variable and independent variables.

CHAPTER 2
Literature Review

Plakandaras et al. (2014) used sophisticated models such as Support Vector Regression and Ensemble Empirical Mode Decomposition to forecast real house prices in the United States using financial and macroeconomic indicators. The models used in this research are good at forecasting national price trends and identifying downturns, but there is still a significant research gap because here they mostly overlook micro-level housing characteristics like lot size, house style, construction quality, number of bathrooms, and garage space.

The study by Vlahovljak (2021) reviews methods such as hedonic pricing models, support vector regression (SVR) and artificial neural networks (ANNs) for predicting house prices and emphasizing the importance of property-level features. The variables considered in the study include location, square footage, number of rooms, property size, and quality of the building.ilding. However, this study focuses mainly on general characteristics and is less concerned with the key structural features of a house.

In the study by Yang (2024), uses the four ensemble learning models, Random Forest, XGBoost, AdaBoost, and Stacking, to forecast US housing prices using a Kaggle dataset including 39,982 properties. The vaiables of this study is number of beds and baths, living space, zip code density, household income, and geographic features. The research uses data techniques such as label encoding and Min-Max scaling, followed by model evaluation using RMSE, R-squared, MAPE, and also explaining the variance score. In here the research gap identified is the less ability of the traditional models used to handle nonlinear relationships and multi-linearity. The models were also implemented using Python tools such as pandas, sklearn, and seaborn.

Plakandaras et al. (2014), Vlahovljak (2021) and Yang (2024) emphasize the use of advanced machine learning models such as Support Vector Regression, Artificial Neural Networks, and ensemble techniques for predicting housing prices showing the trends and complex patterns of how the price changes with different variables. Our case study uses multiple linear regression methods to the variables logarithm of lot area, house style,

overall quality, year built/remodified, basement and ground living area, heating quality, number of bathrooms, number of ground-level rooms, fireplaces, and garage area, and also this approach emphasizes the interpretability and the direct impact of physical housing features on price estimation.

CHAPTER 3
Aims and Objectives

The main objective of this study is to investigate the relationship between selected qualitative and structural characteristics of residential properties and the sale price of residential properties using a multiple linear regression model.

3.1 Specific Objectives

- To identify the relationship between lot size (log-transformed) and house sale price.
- To identify the relationship between overall material and construction quality and house sale price.
- To identify the relationship between year built (centered) and house sale price.
- To identify the relationship between total basement area and house sale price.
- To identify the relationship between above-ground living area and house sale price.
- To identify the relationship between total number of rooms above ground (excluding bathrooms) and house sale price.
- To identify the relationship between garage area and house sale price.

CHAPTER 4
Methodology

4.1 Conceptual Model

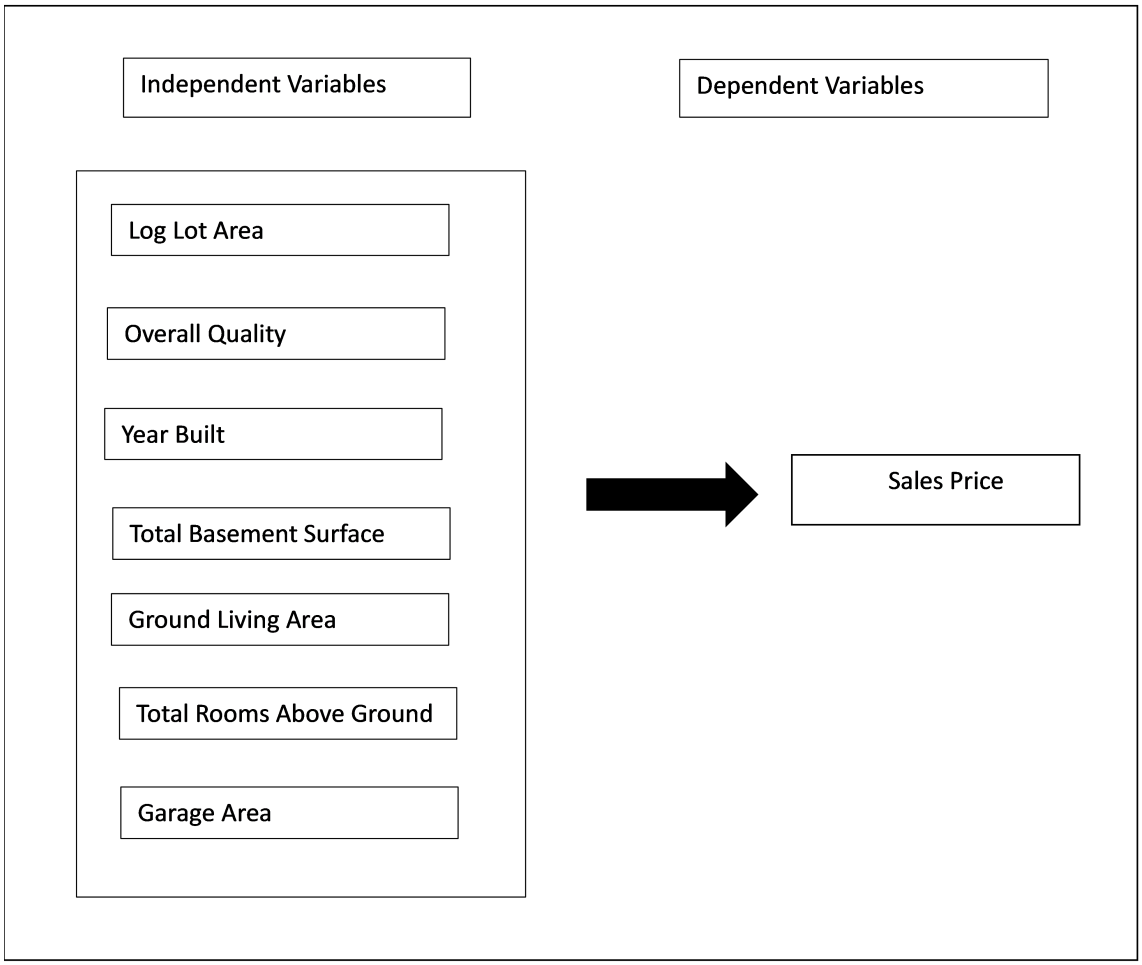


Figure 4.1: Conceptual Model

4.2 Research Approach

The overall research approach for this study is quantitative research approach.Quantitative research is the process of collecting and analyzing numerical data. It can be used to find patterns, to asses relationship between variables, to make predictions, to test hypothesis and generalized result to wider the populations. The multiple linear regression model is one of the tools of the quantitative approach. It used to determine a mathematical relationship among several variables. We have one dependent variable and several independent variables. Therefore, we selected multiple linear regression as our model.

4.2.1 Research Design

The Primary goal of the study to use multiple linear regression to assess how Garage Area, Total Rooms Above Ground, Ground Living Area, Year Built, Total Basement Area, Overall Quality, Lot Area impact on Sales price in Ames Housing Prices,USA.The step wise approach was adopted due to the fact that it allows for the recognition of the most relevant independent variabels.After that, using the most relevant independent variables best fitted regression model is constructed for making the conclusion.

Multiple Linear Regression Model

Multiple linear regression is a statistical technique used to model the relationship between multiple independent variables and a dependent variable. A population model for a multiple linear regression model is written as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

Where:

- Y = Dependent Variable
- $X_1, X_2, \ldots, X_n$ = Independent variables
- β_0 = Y-intercept (the value of Y when all X variables are 0)
- $\beta_1, \beta_2, \ldots, \beta_n$ = Regression coefficients (the change in Y for a one-unit change in the corresponding X variable, holding other X variables constant)
- ϵ = Error term (represents the unobserved factors that affect Y)

CHAPTER 5
Data

Overview

In this chapter, we concern about the data set what we selected. This will gives brief introduction about data set and special characteristic related to data set

5.1 About the Data Set

The data set offers a comprehensive overview of the main factors that influence house prices. It has been compiled to explore how various structural and qualitative features of a property affect its market value.

The R code used to read this data is in appendix.

The data set used in this study is a transformed version of the Kaggle-Ames housing data set. Originally, it consisted of 79 explanatory variables. After removing irrelevant variables and applying suitable transformations, the final dataset comprises 7 independent variables and 1 dependent variable

5.2 Metadata

The source of the data is Kaggle:

`https://www.kaggle.com/datasets/shashanknecrothapa/ames-housing-dataset`

The data collection took place at 2.55 PM on 23 May 2025.

5.3 Data Dictionary

5.3.1 Variable Descriptions

1. **SalePrice** : Sale price of the property (target variable for regression)
2. **log_lot_area** : Natural logarithm of the lot area (in square feet)
3. **Overall Qual** : Overall quality of the house (1 = Very Poor, 10 = Excellent)
4. **NEW_YEAR_BUILT** : A transformed variable representing the year the house was built
5. **Total Bsmt SF** : Total square footage of the basement area
6. **Gr Liv Area** : Above-ground living area in square feet
7. **TotRms AbvGrd** : Total number of rooms above ground (excluding bathrooms)
8. **Garage Area** : Size of the garage in square feet

5.3.2 Variable Types

Variable	**Variable Type**
SalePrice	Numerical
log_lot_area	Numerical (Transformed)
Overall Qual	Ordinal
NEW_YEAR_BUILT	Numerical (Centered)
Total Bsmt SF	Numerical
Gr Liv Area	Numerical
TotRms AbvGrd	Numerical (Discrete)
Garage Area	Numerical

5.4 Dataset Preparation

Check null values and NA values

We checked whether the data set contained null values or `NA` values using R. Then, we remove outliers and rows with null values or empty cells. This cleaning process reduced the data set from 2,930 to 1,196 complete observations.

Remove the variables which which were not mainly affect on housing prices and filtering data

We only use 7 independent variabels for our study which are mainly affect on a housing price.We filtered the data to include only houses with excellent heating quality (`Heating QC = 'Ex'`). This decision was made because the frequencies of other heating quality categories were too small to provide reliable statistical inference and could have introduced noise into the model.

Create New Variables

Some variables were transformed to address skewness and improve linearity:

- **sales price** was transformed using the natural logarithm, resulting in **log_sales**.
- **LotArea** was transformed using the natural logarithm, resulting in **log_lot_area**.
- **YearBuilt** was transformed into **NEW_YEAR_BUILT** using a centering method.

CHAPTER 6
Results

6.1 Exploratory Data Analysis

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
log_lot_area	1196	0	9.0653	0.0129	0.4452	7.5443	8.9227	9.1352	9.3238	10.5811
Overall Qual	1196	0	6.5151	0.0324	1.1191	3.0000	6.0000	7.0000	7.0000	9.0000
NEW_YEAR_BUILT	1196	0	799.3	37.9	1310.0	0.3	210.5	380.6	506.6	9118.6
Total Bsmt SF	1196	0	1083.2	9.60	331.9	173.0	835.0	1040.5	1346.8	1922.0
Gr Liv Area	1196	0	1532.5	11.2	387.3	672.0	1278.5	1509.5	1752.3	2978.0
TotRms AbvGrd	1196	0	6.4574	0.0387	1.3385	3.0000	6.0000	6.0000	7.0000	13.0000
Garage Area	1196	0	510.96	4.59	158.80	160.00	420.00	499.00	590.00	1043.00
SalePrice	1196	0	192263	1404	48551	82500	153500	188000	227000	300000

Figure 6.1: Exploratory data analysis

Sales Price

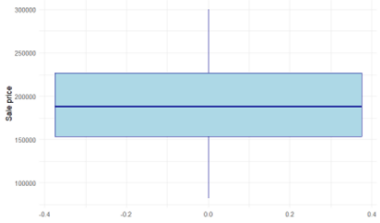


Figure 6.2: Boxplot of Sale Price

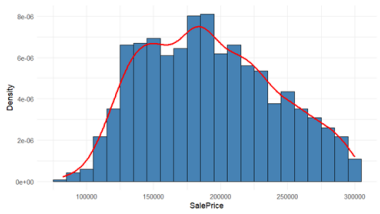


Figure 6.3: Histogram of Sale Price

As in the 'Sale Price' distribution above,It is not normally distributed. It is clearly right-skewed (positively skewed). This means there are a significant number of properties sold at higher prices, pulling the average value upward.The above box plot confirms this spread, with a relatively long upper whisker and number of outliers on the higher end of 'Sale Price.'

Overall Quality and Total rooms

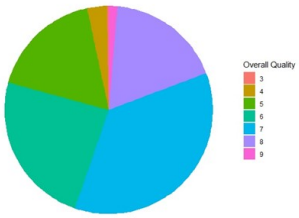


Figure 6.4: Pie chart of Overall Quality

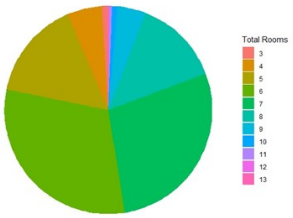


Figure 6.5: Pie chart of total rooms

Log Lot Area

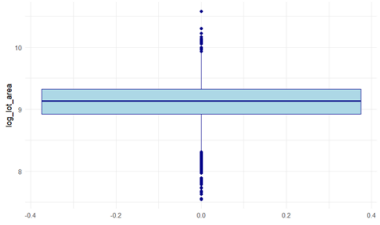


Figure 6.6: Boxplot of Log Lot Area

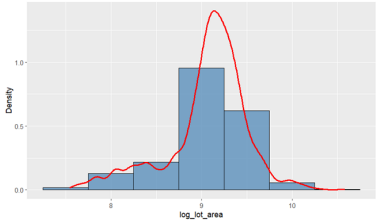


Figure 6.7: Histogram of Log Lot Area

Based on both the histogram and the box plot above, the log lot area variable shows a distribution that is much closer to a normal distribution than the untransformed 'Lot Area'. The transformation has effectively reduced the skewness, resulting in a more symmetric, bell-shaped curve.

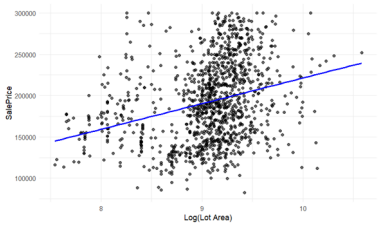


Figure 6.8: Scatter plot of Log Lot Area

The relationship between the logarithmic lot area and the sale price appears to be approximately linear.

Year Built

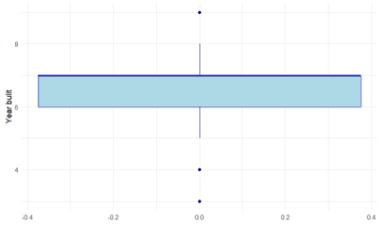


Figure 6.9: Box plot of Year Built

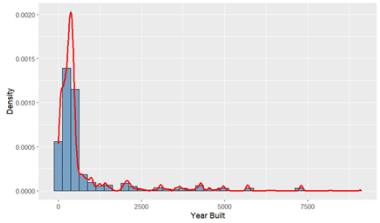


Figure 6.10: Histogram of Year built

By the above histogram and the box plot, the 'Year Built' variable is not normally distributed, and the numerous lower outliers in the box plot all confirm that the 'Year Built' distribution is left-skewed.

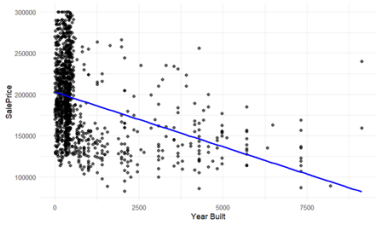


Figure 6.11: Scatter plot of Year built

The relationship appears somewhat linear, but not perfectly. There is a clear positive relationship between 'Year Built' and 'Sale Price.' As the year built increases, the sale price generally tends to be higher.

Total Basement Surface

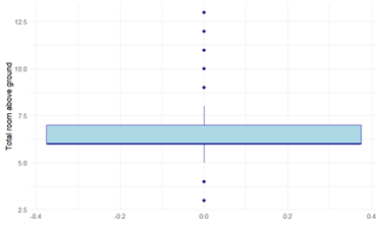


Figure 6.12: Box plot of Total Basement Surface

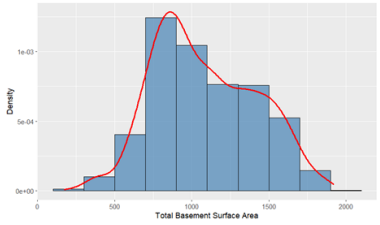


Figure 6.13: Histogram of Total Basement Surface

The above histogram and the box plot show that the 'Total Basement SF' variable is highly skewed and not normally distributed. The presence of many zero values combined with a long, positive tail for non-zero values makes this a very challenging distribution for models that assume normality.

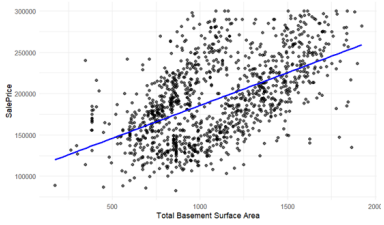


Figure 6.14: Scatter plot of Total Basement Surface

According to the above scatter plot there is an approximately linear positive relationship.

Ground Living Area

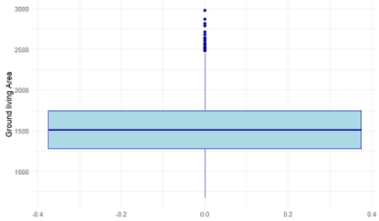


Figure 6.15: Box plot of Ground Living Area

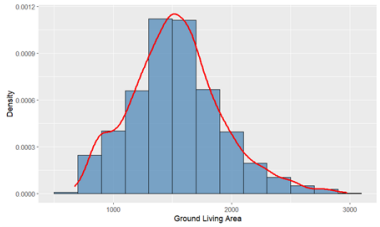


Figure 6.16: Histogram of Ground Living Area

Based on the above two plots "Ground living area" data are right-skewed (positively skewed). The tail extends towards higher living areas, indicating that while most homes have a typical size, there are a number of homes with significantly larger ground living

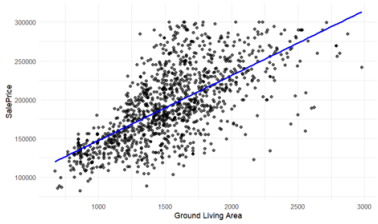


Figure 6.17: Scatter plot of Ground Living Area

area and the 'Gr Living Area' variable is not normally distributed. The above scatter plot shows a positive linear relationship between 'Ground Living Area' and 'Sale Price'. Ground Living Area' is a variable that exhibits a strong and mostly linear relationship with 'Sale Price', making it a valuable feature for linear modeling.

Garage Area

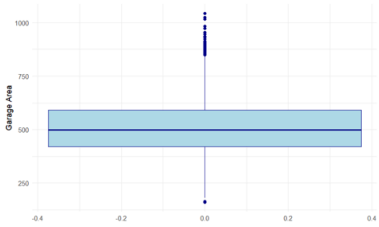


Figure 6.18: Box plot of Garage Area

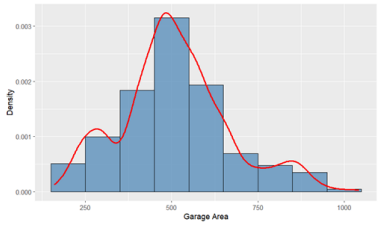


Figure 6.19: Histogram of Garage Area

As above histogram and the box plot The distribution for 'Garage Area' is extremely right-skewed (positively skewed). It has a very long tail extending towards larger garage areas And according to the graphs it is not normally distributed. According to the above

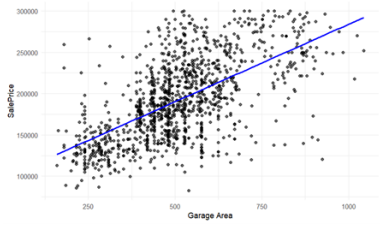


Figure 6.20: Scatter plot of Garage Area

scatter plot there is a general positive trend. As the garage area increases, the 'Sale Price' generally tends to increase.The relationship appears to be roughly linear. A straight line could broadly describe the upward trend, but there's noticeable scatter around this line.

Correlation Plot of Variables

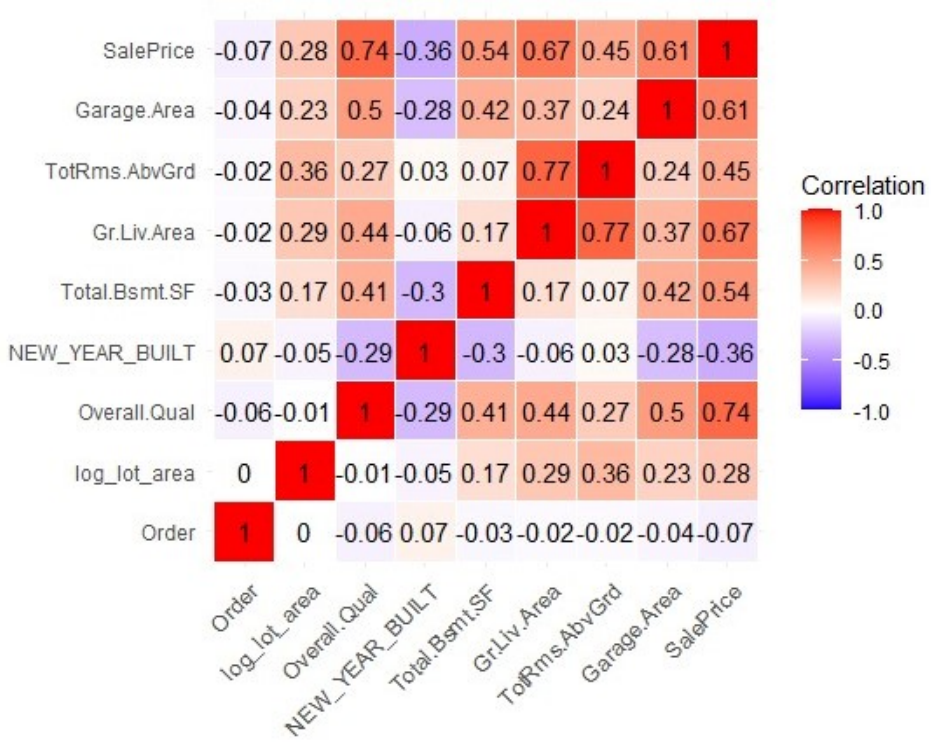


Figure 6.21: Correlation Matrix of Variables

According to this correlation plot,There is a strong positive relationship between Ground Living Area (0.77) and Overall Quality (0.74)with sale price, where larger living areas and higher overall qualities are associated with higher sales prices.There is a moderately positive relationship between garage area,total rooms above ground,Total basement surface and log lot area with sale price.There is a moderate negative relationship between year built and sale price.

6.2 Quantitative Data Analysis

As mentioned earlier, there are no missing values and null values in our data set. In this quantitative data analysis part, we are trying to get the most suitable model for our data set and then we will get the idea about what factors are mainly effected to the sales price of houses.

6.2.1 Correlation Analysis

A correlation table is a matrix that displays the correlation coefficients between multiple variables in a dataset. A positive correlation (closer to 1) indicates that as one variable increases, the other tends to also increase. A negative correlation (closer to-1) indicates that as one variable increases, the other tends to decrease. A correlation close to 0 indicates a weak or no linear relationship. Overall, the correlation table suggests that there are significant positive, negative, and no correlations between all these variables; between the dependent variable (SALES) and all the other independent variables. Therefore, we can make decisions and understand the linear relationship of these variables.

Table: Correlation Table of Numeric Variables

	Order	log_lot_area	NEW_YEAR_BUILT	Total_Bsmt_SF	Gr_Liv_Area	Garage_Area	SalePrice	Log_Sales
Order	1.0000	-0.0012	0.0703	-0.0330	-0.0214	-0.0403	-0.0665	-0.0688
log_lot_area	-0.0012	1.0000	-0.0518	0.1698	0.2874	0.2292	0.2846	0.2822
NEW_YEAR_BUILT	0.0703	-0.0518	1.0000	-0.3043	-0.0617	-0.2772	-0.3574	-0.3899
Total_Bsmt_SF	-0.0330	0.1698	-0.3043	1.0000	0.1700	0.4170	0.5431	0.5359
Gr_Liv_Area	-0.0214	0.2874	-0.0617	0.1700	1.0000	0.3699	0.6674	0.6729
Garage_Area	-0.0403	0.2292	-0.2772	0.4170	0.3699	1.0000	0.6131	0.6117
SalePrice	-0.0665	0.2846	-0.3574	0.5431	0.6674	0.6131	1.0000	0.9898
Log_Sales	-0.0688	0.2822	-0.3899	0.5359	0.6729	0.6117	0.9898	1.0000

Figure 6.22: The Correaltion table

In the above figure we can see there are no linear relationship among the two independent variables. If we have the relationship between two independent variables that is greater than 0.8 we can remove those two independent variables. Then we imagined the correlation between above mentioned two independent variables is insignificant to our study.According to the above table Total.Bsmt.SF(0.5431),Gr.Liv.Area(0.6674) and Garage.Area(0.6131) show a strong positive correlation when larger those area is a strong predictor of a higher sale price.log lot area(0.2846) and NEW YEAR BUILT(-0.3574) shows a weak to moderate positive and weak to moderate negative correlation respectively.

6.2.2 Regression Analysis

As mentioned in above our data set does not have any duplicate value and null values. Then we have tried to get a fitted model for the dependent variable that we have and our data set maybe have some outliers at the moment when we are doing for the fit. As shown in the below box plot and histogram we have deduct outliers.As if our data set have outliers we can not apply the regression to get the best model for our dataset.

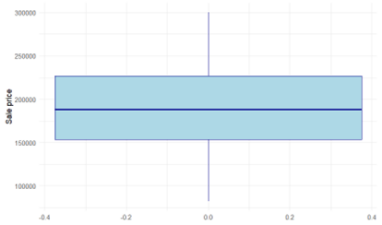


Figure 6.23: Boxplot of Sale Price



Figure 6.24: Histogram of Sale Price

We can not apply regression analyze if we can not satisfy it's assumptions.Then we got the logarithm of dependent variables and the we saw the assumptions are satisfied by that

Those assumptions are,

01. The residual values are normally distributed.

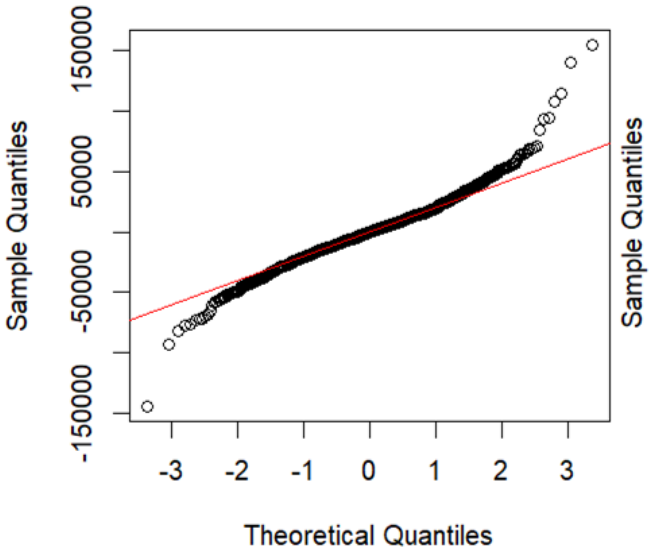


Figure 6.25: QQ plot of the original data

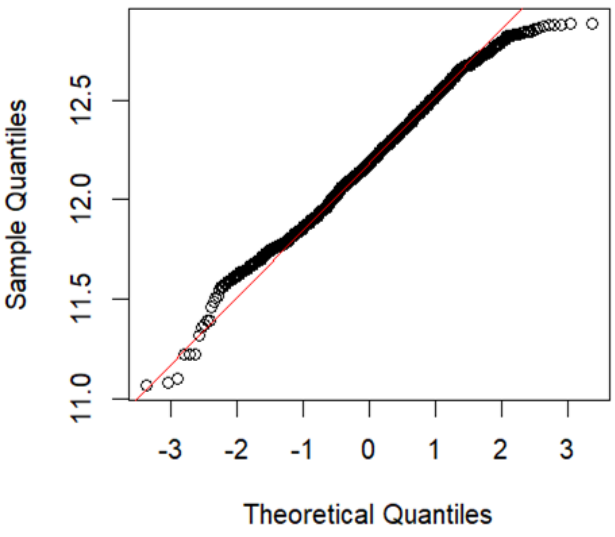


Figure 6.26: QQ plot of logarithm data set

The original QQ plot's points form an "S" shape.The points in the middle are close to the line but the points at the far ends deviate significantly.This indicated Original data is not linear and so not normally distributed.The second plot shows data after logarithmic transformation.Then the points are clustered much more around the line,The plot indicates that the log transformed data has good linearity and is a much better fit for a normal distribution.

02. There must be a linear relationship between the dependent and independent variables

As we have shown scatter plots in the Exploratory Data Analysis part we can see that all independent variables and the dependent variable have a linear relationship. In addition to the correlation, we can also see the relationship between independent and dependent variables. Then our assumption is also satisfied.

03. Multicolinearity

Multicolinearity occurs when two or more independent variables in a regression model are highly correlated. This can be difficult for the model to accurately estimate the individual effect of each variable. In the figure 23 (A matrix of Pearson's correlation), we can see that the independent variables are not highly correlated with each other. Then our assumption is also satisfied.

We used GVIF values to measure multicolinearity where the standard VIF is not sufficient.

	GVIF	Df	GVIF^(1/(2*Df))
Overall.Qual	2.989193	7	1.081355
TotRms.AbvGrd	4.452380	10	1.077531
log_lot_area	1.379213	1	1.174399
NEW_YEAR_BUILT	1.234916	1	1.111268
Total.Bsmt.SF	1.788223	1	1.337245
Gr.Liv.Area	3.687051	1	1.920170
Bedroom.AbvGr	2.401934	1	1.549817
Garage.Area	1.901304	1	1.378878

Figure 6.27: GVIF values

As in the figure 6.27 we used adjusted GVIF values.As they are close to 1 there are no any multicolinearity.

04. Homoscedasticity

It is a key assumption in linear regression that describes a situation where the residual is constant across all level of the independent variables.We checked it by plotting predicted values against the standardized residual values.

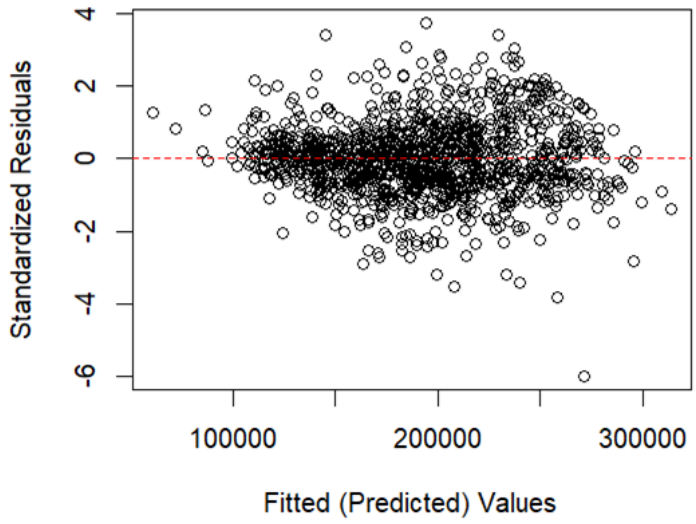


Figure 6.28: Homoscedasticity

There are some outliers in this figure , but considering our large observation we ignored them for our research. By satisfying these assumptions now we can apply multiple linear regression. Before we do the regression analysis, we need to assign the numerical values for our 2 categorical variables and then we will able to enter those variables to our regression model. So that we have used R software encoding function.

Estimated Model Parameters

In this section we are going to build up our model on basis of the relationship between dependent and independent variables.We are going to choose which independent variables to take part in the model by using mathematical models.First we converted our categorical variables to numerical variables using R software.Then got the model as below.

```
Call:
lm(formula = Log_sales ~ Overall_Qual + TotRms_AbvGrd + log_lot_area +
    NEW_YEAR_BUILT + Total_Bsmt_SF + Gr_Liv_Area + Garage_Area,
    data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-0.63880 -0.06395  0.00224  0.06344  0.40321

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.069e+01  9.713e-02 110.022  < 2e-16 ***
Overall_Qual4    2.442e-04  6.542e-02   0.004  0.99702
Overall_Qual5    8.417e-02  6.333e-02   1.329  0.18407
Overall_Qual6    1.836e-01  6.340e-02   2.895  0.00386 **
Overall_Qual7    2.728e-01  6.354e-02   4.293 1.91e-05 ***
Overall_Qual8    3.691e-01  6.417e-02   5.752 1.12e-08 ***
Overall_Qual9    4.893e-01  6.937e-02   7.053 2.98e-12 ***
TotRms_AbvGrd4  -2.700e-02  3.446e-02  -0.784  0.43345
TotRms_AbvGrd5  -3.943e-02  3.310e-02  -1.191  0.23378
TotRms_AbvGrd6  -4.201e-02  3.314e-02  -1.268  0.20512
TotRms_AbvGrd7  -5.676e-02  3.406e-02  -1.667  0.09585 .
TotRms_AbvGrd8  -4.468e-02  3.571e-02  -1.251  0.21115
TotRms_AbvGrd9  -7.075e-02  3.945e-02  -1.793  0.07317 .
TotRms_AbvGrd10 -9.167e-02  5.081e-02  -1.804  0.07142 .
TotRms_AbvGrd11 -1.365e-01  6.738e-02  -2.026  0.04304 *
TotRms_AbvGrd12 -1.006e-01  8.606e-02  -1.169  0.24278
TotRms_AbvGrd13 -1.648e-01  1.155e-01  -1.426  0.15405
log_lot_area     6.716e-02  8.096e-03   8.296 2.93e-16 ***
NEW_YEAR_BUILT  -2.922e-05  2.659e-06 -10.988  < 2e-16 ***
Total_Bsmt_SF    1.478e-04  1.155e-05  12.796  < 2e-16 ***
Gr_Liv_Area      2.729e-04  1.423e-05  19.179  < 2e-16 ***
Garage_Area      1.972e-04  2.517e-05   7.835 1.04e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1087 on 1174 degrees of freedom
Multiple R-squared:  0.8273,    Adjusted R-squared:  0.8242
F-statistic: 267.8 on 21 and 1174 DF,  p-value: < 2.2e-16
```

Figure 6.29: summary of the full model

According to the figure our full model is,

Log sales = 1.069e+01 + 2.442e-04 Overall.Qual4 + 8.417e-02 Overall.Qual5 +1.836e-01 Overall.Qual6 + 2.728e-01 Overall.Qual7 + 3.691e01 Overall.Qual8 + 4.893e-01 Overall.Qual9 - 2.700e-02 TotRms.AbvGrd4 - 3.943e-02 TotRms.AbvGrd5 - 4.201e-02 TotRms.AbvGrd6 - 5.676e-02 TotRms.AbvGrd7- 4.468e-02 TotRms.AbvGrd8 - 7.075e-02 TotRms.AbvGrd9 - 9.167e-02 TotRms.AbvGrd10 - 1.365e-01 TotRms.AbvGrd11 - 1.006e-01 TotRms.AbvGrd12 - 1.648e-01 TotRms.AbvGrd13 + 6.716e-02 log lot area - 2.922e-05 NEW YEAR BUILT + 1.478e-04 Total.Bsmt.SF + 2.729e-04 Gr.Liv.Area + 1.972e-04 Garage.Area

In the figure we can see the R squared value of full model is 0.8273. l

Analysis of Variance Table

Response: Log_sales

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Overall_Qual	6	45.767	7.6278	646.125	< 2.2e-16	***
TotRms_AbvGrd	10	5.905	0.5905	50.018	< 2.2e-16	***
log_lot_area	1	3.249	3.2494	275.247	< 2.2e-16	***
NEW_YEAR_BUILT	1	2.891	2.8912	244.900	< 2.2e-16	***
Total_Bsmt_SF	1	2.978	2.9780	252.258	< 2.2e-16	***
Gr_Liv_Area	1	4.869	4.8690	412.441	< 2.2e-16	***
Garage_Area	1	0.725	0.7248	61.394	1.041e-14	***
Residuals	1174	13.860	0.0118			

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Figure 6.30: ANOVA Table of the full model

Backward Elimination

Backward elimination method is used to fit the most suitable model for this data set.Backward elimination is a step wise variable selection method used in regression analysis to identify the most relevant independent variables that should be included in a regression model. It is a technique for simplifying the model while retaining its predictive power.Multiple linear regression usually uses backward elimination.

The process is iterative and works by starting with all available features and then removing them one by one based on their statistical significance. Here we used the significance level as 0.01 for our study.According to the theorem we used p-value,R squared value,AIC values to select the best model.The steps involved in the regression model's backward elimination procedure are shown below figure. We used R software for our analysis.

```
Backward Elimination Method
---------------------------

Candidate Terms:

1. Overall_Qual
2. TotRms_AbvGrd
3. log_lot_area
4. NEW_YEAR_BUILT
5. Total_Bsmt_SF
6. Gr_Liv_Area
7. Garage_Area


Step   => 0
Model  => Log_sales ~ Overall_Qual + TotRms_AbvGrd + log_lot_area + NEW_YEAR_BUILT + Total_Bsmt_SF + Gr_Liv_Area + Garage_Area
R2     => 0.827

Initiating stepwise selection...

Step     => 1
Removed  => TotRms_AbvGrd
Model    => Log_sales ~ Overall_Qual + log_lot_area + NEW_YEAR_BUILT + Total_Bsmt_SF + Gr_Liv_Area + Garage_Area
R2       => 0.82574


No more variables to be removed.

Variables Removed:

=> TotRms_AbvGrd
```

Stepwise Summary

Step	Variable	AIC	SBC	SBIC	R2	Adj. R2
0	Full Model	-1891.382	-1774.387	-5313.184	0.82728	0.82419
1	TotRms_AbvGrd	-1900.767	-1834.640	-5304.732	0.82574	0.82412

Final Model Output

Model Summary

R	0.909	RMSE	0.108
R-Squared	0.826	MSE	0.012
Adj. R-Squared	0.824	Coef. Var	0.896
Pred R-Squared	0.822	AIC	-1900.767
MAE	0.081	SBC	-1834.640

RMSE: Root Mean Square Error
MSE: Mean Square Error
MAE: Mean Absolute Error
AIC: Akaike Information Criteria
SBC: Schwarz Bayesian Criteria

ANOVA

	Sum of Squares	DF	Mean Square	F	Sig.
Regression	66.260	11	6.024	510.045	0.0000
Residual	13.983	1184	0.012		
Total	80.244	1195			

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AIC: Akaike Information Criteria
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Regression	66.260	11	6.024	510.045	0.0000
Residual	13.983	1184	0.012		
Total	80.244	1195			

Parameter Estimates

model	Beta	Std. Error	Std. Beta	t	Sig	lower	upper
(Intercept)	10.692	0.091		117.856	0.000	10.514	10.870
Overall_Qual4	0.007	0.065	0.004	0.103	0.918	-0.121	0.135
Overall_Qual5	0.086	0.063	0.126	1.359	0.174	-0.038	0.210
Overall_Qual6	0.189	0.063	0.311	2.982	0.003	0.065	0.313
Overall_Qual7	0.277	0.063	0.514	4.372	0.000	0.153	0.402
Overall_Qual8	0.373	0.064	0.552	5.825	0.000	0.248	0.499
Overall_Qual9	0.495	0.069	0.219	7.139	0.000	0.359	0.630
log_lot_area	0.064	0.008	0.111	8.255	0.000	0.049	0.080
NEW_YEAR_BUILT	0.000	0.000	-0.152	-11.413	0.000	0.000	0.000
Total_Bsmt_SF	0.000	0.000	0.189	13.053	0.000	0.000	0.000
Gr_Liv_Area	0.000	0.000	0.377	25.565	0.000	0.000	0.000
Garage_Area	0.000	0.000	0.122	7.937	0.000	0.000	0.000

Figure 6.31: Reduced model

In the full model summary from the the R-Squared value is 0.82728 and the Adjusted R-Squared value is 0.82419.This indicates that the independent variables in the model explained about 82.7 percent of the variance in the dependent variable.After removed the variable TotRms.AbvGrd the R squared became 0.82574 and Adj.R squared become 0.82412.

Our reduced model is,

Log space = 10.692 + 0.007 Overall.Qual4 + 0.086 Overall.Qual5 + 0.189 Overall.Qual6 + 0.277 Overall.Qual7 + 0.373 Overall.Qual8 + 0.495 Overall.Qual9 + 0.064 log lot area

Below shows the comparison of full model and the reduced model,

```
Analysis of Variance Table

Model 1: Log_sales ~ Overall_Qual + log_lot_area + NEW_YEAR_BUILT + Total_Bsmt_SF +
    Gr_Liv_Area + Garage_Area
Model 2: Log_sales ~ Overall_Qual + TotRms_AbvGrd + log_lot_area + NEW_YEAR_BUILT +
    Total_Bsmt_SF + Gr_Liv_Area + Garage_Area
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1   1189 14.037
2   1174 13.860 15   0.17704 0.9998 0.4526
```

Figure 6.32: Anova Table

Hypothesis Testing

Hypothesis 1

- Null Hypothesis(H0)

$$\beta(i) = 0, i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 \tag{6.1}$$

There is no linear relationship between dependent variable and independent variables.

- Alternative Hypothesis(H1

$$\beta(i) \neq 0, i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 \tag{6.2}$$

There is a linear relationship between dependent variable and independent variables. According to the 6.32 I did the following hypothesis testing.

1. Relationship between Overall Quality 4 and log sales price

   t value is 0.103 and we can see that p value is greater than 0.01 for overall quality.As it is a dummy varible we reject H0.

$$\beta(1) \neq 0 \tag{6.3}$$

$$\beta(1) = 0.007 \tag{6.4}$$

2. Relationship between Overall Quality 5 and log sales price

   t value is 1.359 and we can see that p value is greater than 0.01 for overall quality.As it is a dummy varaible we rejecT H0.

$$\beta(1) \neq 0 \tag{6.5}$$

$$\beta(1) = 0.0.86 \tag{6.6}$$

3. Relationship between Overall Quality 6 and log sales price

   t value is 2.982 and we can see that p value is very small than 0.01 for overall quality.Then there is a significant relationship between these two.Then we can reject the H0.

$$\beta(1) \neq 0 \tag{6.7}$$

$$\beta(1) = 0.189 \tag{6.8}$$

4. Relationship between Overall Quality 7 and log sales price

   t value is 4.372 and we can see that p value is very small than 0.01 for overall quality.Then there is a significant relationship between these two.Then we can reject the H0.

$$\beta(1) \neq 0 \tag{6.9}$$

$$\beta(1) = 0.277 \tag{6.10}$$

5. Relationship between Overall Quality 8 and log sales price

   t value is 5.825 and we can see that p value is very small than 0.01 for overall quality.Then there is a significant relationship between these two.Then we can reject the H0.

$$\beta(1) \neq 0 \tag{6.11}$$

$$\beta(1) = 0.373 \tag{6.12}$$

6. Relationship between Overall Quality 9 and log sales price

   t value is 7.139 and we can see that p value is very small than 0.01 for overall quality.Then there is a significant relationship between these two.Then we can reject the H0.

$$\beta(1) \neq 0 \tag{6.13}$$

$$\beta(1) = 0.495 \tag{6.14}$$

7. Relationship between log lot area and log sales price

   t value is 8.255 and we can see that p value is very small than 0.01 for overall quality.Then there is a significant relationship between these two.Then we can reject the H0.

$$\beta(1) \neq 0 \tag{6.15}$$

$$\beta(1) = 0.064 \tag{6.16}$$

8. Relationship between NEW YEAR BUILT and log sales price

   t value is -10.348 .we can not reject the H0.Then that variable has not any significance relationship with the dependent variable.It is not important for the model.

$$\beta(1) = 0.000 \tag{6.17}$$

9. Relationship between Total basement surface and log sales price

   t value is 13.131 and we can see that p value is very small than 0.01 for overall quality.Then that variable has not any significance relationship with the dependent variable.It is not important for the model.

$$\beta(1) = 0.000 \tag{6.18}$$

10. Relationship between Ground living area and log sales price

    t value is 26.292 and we can see that p value is very small than 0.01 for overall quality.that variable has not any significance relationship with the dependent variable.It is not important for the model.

$$\beta(1) = 0.000 \tag{6.19}$$

11. Relationship between Garage Area and log sales price

    t value is 8.189 and we can see that p value is very small than 0.01 for overall quality.That variable has not any significance relationship with the dependent variable.It is not important for the model.

$$\beta(1) = -0.000 \tag{6.20}$$

Hypothesis 2

• Null hypothesis (H0):

Reduced model is suitable

• Alternative hypothesis (Ha):

Full model is suitable

First, we should consider first the F partial.

$$F = \frac{(SSE_{reduced} - SSE_{full})/(p-m)}{SSE_{full}/(n-p-1)}$$

$$= \frac{(14.037 - 13.860)/(21-6)}{(13.860)/(1196-21-1)}$$

$$= 0.9998$$

Then we get the F table value. It for 99 percent confidence interval is,

$$F(1-0.01, 15, 1174) = 2.053826$$

The F partial $<$ F table, i.e $0.9998 < 2.05382$

Then we do not reject null hypothesis at 0.01 significance level. Therefore,

The reduced model is suitable.

Model Interpretation

1. Intercept

   When all the predictors are equal to zero the predicted value of the log sales is increase by 10.692.

2. Overall_Qual4

   A house with the Overall_Qual4 is predicted to have a price that is 0.007 greater than the house of base quality while all other variables are constant.

3. Overall_Qual5

   A house with the Overall_Qual5 is predicted to have a price that is 0.086 greater than the house of base quality while all other variables are constant.

4. Overall_Qual6

   A house with the Overall_Qual6 is predicted to have a price that is 0.189 greater than the house of base quality while all other variables are constant.

5. Overall_Qual7

   A house with the Overall_Qual7 is predicted to have a price that is 0.277 greater than the house of base quality while all other variables are constant.

6. Overall_Qual8

   A house with the Overall_Qual8 is predicted to have a price that is 0.373 greater than the house of base quality while all other variables are constant.

7. Overall_Qual9

   A house with the Overall_Qual9 is predicted to have a price that is 0.495 greater than the house of base quality while all other variables are constant.

8. log_lot_area

   The change of expected log sales for one unit of increasement in log_lot_area is 0.064 while all other variables are constant.

CHAPTER 7
Discussion and Conclusion

Discussion

Our study is aimed to predict housing prices using independent variables such as log lot area,Overall quality,Year built,Total basement surface,Ground living area,Total room above ground,Garage area.We began with Descriptive analysis,visualizing relationship through plots and summary statistics.This exploratory highlighted potential predictors but also revealed the complexity of these relationships.We ensured residuals were normally distributed for valid statistical inference. Multicollinearity was carefully examined to prevent errors in model estimates, and we confirmed homoscedasticity to maintain consistent error variance across predictors.

Initially we tried a full model with all variables but found it is unstable due to potential overfitting and redundancy among the predictors.To refine our model we used Backward elimination regression technique to assess variable significance systematically.As in the figure 6.31 By the R software we get the reduced model.In here model was reduced under the consideration of R squared values,AIC values.

The process starts with the full model that include seven independent variables as described in the previous.We got that the R value of the full model is 0.82728.This indicates approximately 82.7 percent of the variance in the dependent variable (Log_sales)can be explained by the independent variables.The Akaike Information Criterion (AIC) is -1891.382 and the Schwarz Bayseian Criteria (SBC) is -1774.387.The lower values of AIC and BIC values are generally better as they indicate a good balance between model fit and complexity.

In the next step model evaluate TotRms.AbvGrd was the least significant and removed it.As it has the highest p values as in the figure 6.29.Also the Overall.Qual4 has a p value of 0.918 and Overall.Qual5 has a p value of0.174,but they were not removed in the first step it was kept in the final model.After removing it new model is formed with the remaining six variables.The R squared value of new model is 0.82574.As it is close to original R squared value 0.82419 it means removal of TotRms.AbvGrd had a negligible impact on the model.The

AIC improved to -1900.767 and SBC improved to -1834.640.Both values are lower than the full model's,indicated the new model was a better fit.In the final model according to hypothesis testing we remove NEW_YEAR_BUILT,TOTAL_Bsmt.SF,,Gr.Liv.Area and Garage.Area as they had insignificant relationship with the dependent variable.

Conclusion

This study successfully investigated the key factors influencing housing price in Ames,Lowa using multiple linear regression model.The primary objective was to develop a predictive model by identifying a significant subset of variables from a comprehensive dataset.By employing a robust methodology that included data cleaning,variable transformation and a backward elimination approach,a refined model was achieved.

The final analysis confirmed that a select number of variables have a statistically significant relationships with residential property prices.Specially,the model demonstrated that the overall quality is a strong positive predictor of house values.The backward elimination process led to the removal of several variables,including NEW_YEAR_BUILT,Total_Bsmt_SF,Gr_Liv_Area and Garage_Area as their relationship with the dependent variable was found to be insignificant in the final model.The removal of TotRms_AbvGrd was also found to have a negligible impact,resulting in a better model without sacrificing predictive power.

The result provide valuable insights for various stakeholders including prospective homeowners,real state agents and property developers,enabling them to make more inform decisions by focusing on the most influential characteristics of a property.

References

[1] G. E. Sardianou N. Plakandaras and C. T. Plakandaras. Predicting real house prices in the united states using financial and macroeconomic indicators, 2014.

[2] D. Vlahovljak. House price prediction using machine learning models, 2021.

[3] P. Yang. Forecasting u.s. housing prices using ensemble learning models. 2024.

CHAPTER 8
C codes

8.1 R codes for data analyzing part

```
#install and load packages
install.packages("car")
library(dplyr)
library(knitr)
library(car)
library(olsrr)

#Read the data set
data<-read.csv(file="data6.csv")
df<-data.frame(data)

#Convert Categorical variables to numerical
df$Overall_Qual<- as.factor(df$Overall_Qual)
df$ TotRms_AbvGrd<- as.factor(df$TotRms_AbvGrd)

#Fit the full model
model <- lm( SalePrice~Overall_Qual+TotRms_AbvGrd+log_lot_area+
   NEW_YEAR_BUILT+Total_Bsmt_SF+Gr_Liv_Area+Garage_Area , data = df)
summary(model)

#Get the Correlation table
numeric_data <- df %>%
  select_if(is.numeric)
correlation_matrix <- round(cor(numeric_data),4)
kable(correlation_matrix,
      caption = "Correlation Table of Numeric Variables")
summary_data <- summary(df)
summary_data


#residual plot
```

```r
plot(model, which = 1)

#q-q plot for normality
qqnorm(residuals(model))
qqline(residuals(model),col="red")

dependent_variable <- df$SalePrice
hist(dependent_variable,
     main = "Histogram of SalePrice",
     xlab = "SalePrice",
     col = "skyblue",
     border = "white")

#Homoscedasticity
# Get the standardized residuals from the model
standardized_residuals <- rstandard(model)

# Get the fitted (predicted) values from the model
fitted_values <- fitted(model)

# Plot the standardized residuals against the fitted values
plot(fitted_values, standardized_residuals,
     main = "Standardized Residuals vs Fitted Values",
     xlab = "Fitted (Predicted) Values",
     ylab = "Standardized Residuals")

# Add a horizontal line at y=0 for reference
abline(h = 0, col = "red", lty = 2)

#multicolinearity
vif_values<-vif(model)
vif_values
vif_df <- data.frame(Variable = names(vif_values),VIF = vif_values)
ggplot(vif_df, aes(x = reorder(Variable, VIF), y = VIF)) +
  geom_bar(stat = "identity", fill = "steelblue") +
```

```
  geom_hline(yintercept = 5, color = "red", linetype = "dashed") +
  geom_hline(yintercept = 10, color = "darkred", linetype = "dotted") +
  coord_flip() +
  labs(title = "Variance Inflation Factor (VIF) for Predictors",
       x = "Variables",
       y = "VIF Value") +
  theme_minimal()

#full model
model <- lm( Log_sales~Overall_Qual +TotRms_AbvGrd+log_lot_area+
   NEW_YEAR_BUILT+Total_Bsmt_SF+Gr_Liv_Area+Garage_Area , data = df)
summary(model)

#Anova table of the full model
anova(model)

#backward elimination
ols_step_backward_p(model,prem=0.01,details=T)
el.forward<-ols_step_backward_p(model,prem=0.01,details=F)
el.forward

#Reduced model
Reduced_model<-lm(Log_sales ~ Overall_Qual + log_lot_area +
   NEW_YEAR_BUILT + Total_Bsmt_SF + Gr_Liv_Area + Garage_Area  ,data=
   data )
summary(Reduced_model)

#Anova table for comparing full model and reduced model
anova(Reduced_model,model)

#mean square error of full model
MSE_full<-deviance(model)/df.residual(model)
MSE_full

#Degree of freedom for F test
```

```
cmp<-anova(Reduced_model,model)
df1<-cmp$Df[2]
df1
df2<-df.residual(model)
df2

#Critical value
alpha<-0.01
f_critical<-qf(1-alpha,df1,df2)
f_critical
```